

Harshana Lakshara Fernando

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SUMMARY

Full-Stack Software Engineer experienced in end-to-end application delivery, legacy integrations, and production support. Strong in **Java**, **Spring Boot**, **Kafka**, **React**, and DevOps workflows. Known for rapid learning, taking ownership of distributed systems, and choosing solutions that balance reliability, scalability, and real product needs.

EDUCATION

B.Sc. (Hons) in Engineering — Computer Science and Engineering, University of Moratuwa 2022 – 2026
G.C.E. A/L: 3 A's, Physical Science Stream, District Rank 17

EXPERIENCE

R&D Software Engineer — KōreroCare Platform Mar 2026 – Present
Empathic Computing Laboratory, University of Auckland Remote

- Co-led backend design for the KōreroCare mobile-wearable sensing platform, building Go/Gin-based services for time-series ingestion and cross-device synchronization.
- Designed PostgreSQL schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated Keycloak OAuth2/OIDC with Traefik gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an HL7/FHIR-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern — ISA Aviation Technology Dec 2024 – May 2025
Information Systems Associates (Pvt) Ltd Colombo, Sri Lanka

- Owned design and delivery of a distributed Spring Boot/Kafka-backed PNR ID generation service, live in production, achieving 25+ PNRs/sec at <100ms latency.
- Designed a pool-based PNR allocation model to replace legacy Oracle-sequence generation, supporting short domain-specific PNR formats while avoiding predictable sequential identifiers.
- Implemented concurrency-safe PNR allocation using a Kafka-backed pool, ensuring the same PNR was not assigned to multiple service instances, while using Java ReentrantLock for application-level critical sections.
- Led the rollout from legacy PNR generation to AeroOrder using database-driven feature toggles and a strangler-pattern migration approach for phased production cutover and fallback.
- Resolved production Kafka consumer lag and offset-reset issues by adding graceful shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
- Integrated gRPC clients and Kafka consumers into the legacy Java 8 AeroMart monolith for backward-compatible communication with AeroOrder microservices.
- Built async load-testing and monitoring support using JMeter, Prometheus, and Grafana to validate PNR/order flows, queue time, consumer behavior, and service latency.
- Added Jenkins environment support, maintained >90% unit test coverage, and coordinated delivery with QA and distributed engineering teams.

PROJECTS

- Supply Chain DApp Developed a full-stack supply chain tracker with trustless escrow. Solidity, Next.js, Polygon
- Technologies: Solidity, Next.js
- Governance DAO Engineered a decentralized autonomous organization for voting. Solidity, Hardhat
- Technologies: Solidity, OpenZeppelin
- ZK-Rollups Scalability Researched batching and throughput enhancement on Ethereum Layer 2. Circom, Solidity
- Technologies: Circom, Solidity
- Fintech & Crypto Lab Group project focusing on decentralized finance mechanisms. Research
- Technologies: DeFi, Blockchain

SKILLS

Backend: Java, Spring Boot, Go/Gin, Python, Node.js, REST, gRPC, GraphQL, Hibernate
Frontend: React, Next.js, TypeScript, JavaScript, React Query
Messaging & APIs: Kafka, RabbitMQ, WebSockets, Protobuf, JWT, OAuth2/OIDC
Data: PostgreSQL, MySQL, Oracle DB, Redis, Couchbase, DynamoDB, Prisma
Cloud / DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux
Observability / Testing: Prometheus, Grafana, JMeter, Gatling, JUnit, Mockito, Pytest